Unit–3

Concept of Quantum Tunnelling and its applications

List of Questions

A. Short Answer Questions

1. What is meant by a finite potential barrier?
2. Define reflection and transmission coefficients.
3. What is quantum tunnelling?
4. Why is quantum tunnelling not possible in classical mechanics?
5. What is meant by particle flux?
6. Write the expression for particle flux for a plane wave.
7. What is the physical meaning of the reflection coefficient?
8. What is the physical meaning of the transmission coefficient?
9. What is Scanning Tunnelling Microscope (STM)?
10. Why does the tunnelling current in STM change rapidly with distance?
11. Why is alpha decay not possible according to classical physics?
12. Explain alpha decay as a tunnelling phenomenon.
13. Why is STM capable of atomic resolution?

B. Medium Answer Questions

14. Discuss the dependence of tunneling probability on barrier height, width, and particle energy.
15. Explain the concept of particle flux in quantum mechanics. Derive the expression for particle flux for a plane wave and discuss its physical significance.
16. Find the expectation value of position for the ground state of a quantum harmonic oscillator.
17. Explain the working principle of scanning tunnelling microscope**.**
18. Explain the process of alpha decay using quantum tunnelling**.**
19. Discuss the role of tunnelling probability in nuclear decay and nanotechnology.
20. Explain why the ground state energy of the quantum harmonic oscillator is non-zero. What is this energy called and why does it exist?
21. Explain "the half-life of polonium that emits 8-MeV α-particles is only hundreds of milliseconds and the half-life of uranium that emits 4-MeV α-particles is billions of years" by applying quantum mechanical tunnelling.

C. Long Answer Questions

22. Obtain the expression for the transmission coefficient of a particle tunneling through a finite potential barrier using boundary conditions.
23. Describe the behaviour of quantum particles at a potential step using the concepts of particle flux, reflection coefficient, and transmission coefficient.
24. Describe the construction and working of scanning tunneling microscope (STM) with appropriate diagram.

D. Numerical

1. Calculate the tunneling probability for an electron of total energy 0.8 eV incident upon a rectangular potential barrier of height 1.2 eV and width 0.4 nm.

2. An electron of energy E = 5eV is incident on a potential barrier of height U_0 = 10eV and width L = 0.5nm. Calculate the transmission coefficient.

3. *Alpha Decay (Tunnelling Probability):*

   An alpha particle of energy E = 5MeV encounters a nuclear potential barrier of height U_0 = 25MeV and width $L = 2 \times 10^{-14}$ m. Estimate the tunnelling probability.

4. *Scanning Tunnelling Microscope:*

   In an STM, the work function of the metal is ϕ = 4eV. Calculate the ratio of tunnelling currents when the tip-sample distance changes from d_1 = 0.5nm to d_2 = 0.6nm.

5. *Energy Levels:*

   Calculate the first three energy levels (n=0,1,2) of a harmonic oscillator with angular frequency $\omega = 2.0 \times 10^{14}$ rad/s. Express your answers in electron volts (eV).

   Hint: $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$, $\hbar = 1.0546 \times 10^{-34}\,\text{J·s} = 6.582 \times 10^{-16}\,\text{eV·s}$.

6. *Transition Energy:*

   An electron in a quantum harmonic oscillator undergoes a transition from n = 3 to n = 1. If the emitted photon has wavelength λ = 400nm, what is the oscillator frequency ω?

   Hint: $E_{\text{photon}} = \hbar\omega(n_2 - n_1) = \frac{hc}{\lambda}$.

7. *Expectation Value:*

   Show that for the ground state of a harmonic oscillator, $\langle x^2 \rangle = \frac{\hbar}{2m\omega}$. Calculate its numerical value for an electron ($m = 9.11 \times 10^{-31}$ kg) with $\omega = 1.0 \times 10^{15}$ rad/s.

8. Calculate the de Broglie wavelength of the orbital electron in the first orbit of hydrogen atom.

9. *Hydrogen Atom Energy Calculation:*

   Calculate the energy of an electron in the n = 2 state of a hydrogen atom in eV. Also calculate the energy required to ionize a hydrogen atom from the n = 2 state.

   Hint: $E_n = -\frac{13.6}{n^2}$ eV.

10. *Hydrogen Atom Photon Emission:*

    A hydrogen atom transitions from n = 4 to n = 2.

    i. Calculate the energy of the emitted photon in eV.
    ii. Calculate the wavelength of this photon in nm (visible, UV, or IR?).

    Hint: $E = 13.6\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$ eV, $\lambda = \frac{hc}{E}$.